ANSWER KEY

English Test

1. D	21. B	41. C	61. D
2. F	22. F	42. G	62. J
3. B	23. A	43. A	63. B
4. F	24. G	44. J	64. F
5. C	25. A	45. A	65. C
6. J	26. G	46. J	66. G
7. C	27. D	47. D	67. B
8. G	28. G	48. F	68. H
9. A	29. A	49. A	69. D
10. H	30. J	50. J	70. G
11. A	31. D	51. B	71. B
12. J	32. F	52. H	72. G
13. B	33. A	53. D	73. D
14. H	34. G	54. F	74. F
15. D	35. B	55. D	75. B
16. F	36. H	56. G	
17. B	37. D	57. D	
18. H	38. F	58. H	
19. D	39. B	59. A	
20. F	40. H	60. J	

Mathematics Test

1. A	21. E	41. A
2. H	22. J	42. H
3. D	23. C	43. C
4. F	24. G	44. J
5. C	25. B	45. B
6. H	26. G	46. H
7. D	27. D	47. A
8. J	28. F	48. G
9. A	29. B	49. E
10. H	30. H	50. H
11. C	31. C	51. C
12. K	32. F	52. H
13. B	33. A	53. B
14. H	34. J	54. F
15. B	35. C	55. A
16. H	36. K	56. J
17. B	37. E	57. B
18. G	38. J	58. J
19. A	39. E	59. C
20. K	40. K	60. J

Reading Test

1. C	21. B
2. J	22. H
3. B	23. B
4. G	24. G
5. D	25. B
6. G	26. J
7. D	27. A
8. H	28. H
9. A	29. A
10. H	30. J
11. B	31. C
12. H	32. G
13. D	33. B
14. H	34. J
15. B	35. C
16. G	36. F
17. C	37. B
18. J	38. G
19. A	39. D
20. F	40. J

Science Reasoning Test

1. D	21. D
2. J	22. J
3. B	23. C
4. H	24. G
5. D	25. A
6. G	26. H
7. D	27. B
8. H	28. J
9. A	29. C
10. F	30. G
11. B	31. C
12. H	32. J
13. C	33. A
14. G	34. J
15. C	35. C
16. G	36. G
17. D	37. B
18. J	38. H
19. B	39. A
20. H	40. G

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SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

Step 1 Determine your RAW SCORE for each of the four multiple-choice sections:

English _____

Mathematics _____

Reading _____

Science Reasoning _____

The following Raw Score Table shows the total possible points for each section.

RAW SCORE TABLE	
KNOWLEDGE AND SKILL AREAS	**RAW SCORES**
ENGLISH	75
MATHEMATICS	60
READING	40
SCIENCE REASONING	40
WRITING	12

Multiple-Choice Scoring Worksheet

Step 2 Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

English	_____ RAW SCORE	× **36** =	_____	÷ **75** =	_____	**− 2**	(*correction factor)
					_____ SCALE SCORE		
Mathematics	_____ RAW SCORE	× **36** =	_____	÷ **60** =	_____	**+ 1**	(*correction factor)
					_____ SCALE SCORE		
Reading	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 2**	(*correction factor)
					_____ SCALE SCORE		
Science Reasoning	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 1.5**	(*correction factor)
					_____ SCALE SCORE		

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

Step 3 Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

English _____

Writing _____

English/Writing _____

ENGLISH/WRITING SCORING TABLE

ENGLISH SCALE SCORE	WRITING TEST SCORE											ENGLISH SCALE SCORE
	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	
36	26	27	28	29	30	31	32	33	34	32	36	**36**
35	26	27	28	29	30	31	31	32	33	34	35	**35**
34	25	26	27	28	29	30	31	32	33	34	35	**34**
33	24	25	26	27	28	29	30	31	32	33	34	**33**
32	24	25	25	26	27	28	29	30	31	32	33	**32**
31	23	24	25	26	27	28	29	30	30	31	32	**31**
30	22	23	24	25	26	27	28	29	30	31	32	**30**
29	21	22	23	24	25	26	27	28	29	30	31	**29**
28	21	22	23	24	24	25	26	27	28	29	30	**28**
27	20	21	22	23	24	25	26	27	28	28	29	**27**
26	19	20	21	22	23	24	25	26	27	28	29	**26**
25	18	19	20	21	22	23	24	25	26	27	28	**25**
24	18	19	20	21	22	23	23	24	25	26	27	**24**
23	17	18	19	20	21	22	23	24	25	26	27	**23**
22	16	17	18	19	20	21	22	23	24	25	26	**22**
21	16	17	17	18	19	20	21	22	23	24	25	**21**
20	15	16	17	18	19	20	21	21	22	23	24	**20**
19	14	15	16	17	18	19	20	21	22	23	24	**19**
18	13	14	15	16	17	18	19	20	21	22	23	**18**
17	13	14	15	16	16	17	18	19	20	21	22	**17**
16	12	13	14	15	16	17	18	19	20	20	21	**16**
15	11	12	13	14	15	16	17	18	19	20	21	**15**
14	10	11	12	13	14	15	16	17	18	19	20	**14**
13	10	11	12	13	14	14	15	16	17	18	19	**13**
12	9	10	11	12	13	14	15	16	17	18	19	**12**
11	8	9	10	11	12	13	14	15	16	17	18	**11**
10	8	9	9	10	11	12	13	14	15	16	17	**10**
9	7	8	9	10	11	12	13	13	14	15	16	**9**
8	6	7	8	9	10	11	12	13	14	15	16	**8**
7	5	6	7	8	9	10	11	12	13	14	15	**7**
6	5	6	7	7	8	9	10	11	12	13	14	**6**
5	4	5	6	7	8	9	10	11	12	12	13	**5**
4	3	4	5	6	7	8	9	10	11	12	13	**4**
3	2	3	4	5	6	7	8	9	10	11	12	**3**
2	2	3	4	5	6	6	7	8	9	10	11	**2**
1	1	2	3	4	5	6	7	8	9	10	11	**1**

Step 4 Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32.

_____	+	_____	+	_____	+	_____	=	_____
ENGLISH *OR* ENGLISH/WRITING SCALE SCORE		MATHEMATICS SCALE SCORE		READING SCALE SCORE		SCIENCE SCALE SCORE		SCALE SCORE TOTAL

_____	÷ 4 =	_____
SCALE SCORE TOTAL		COMPOSITE SCORE

ANSWERS AND EXPLANATIONS

English Test Explanations

PASSAGE I

1. **The best answer is D.** The sentence is clearest and most concise when the word *back* is inserted in place of the underlined portion. Answer choices A, B, and C are either awkward or redundant.

2. **The best answer is F.** This sentence is best as written. The word *fairly* is correctly used as an adverb to further modify the adjective *regular*, which modifies the noun *trips*. Answer choice G is not correct because the word *fair* is being used as an adjective to describe the trips, when it should be used to modify the adjective *regular*. Answer choice H is incorrect because it reverses the words, thereby changing the meaning of the sentence. It is not best to use *regular* because the context of the sentence indicates that the trips were *mostly* between Las Vegas and New York, so answer choice J is incorrect.

3. **The best answer is B.** The context of the sentence indicates that first the author goes to Las Vegas, "then" the author returns home to New York. It is not appropriate to use "than" to indicate a progression in time, so answer choices A and C are incorrect. Answer choice D is wordy and is not as clear and effective as answer choice B.

4. **The best answer is F.** The sentence is best as it is written. It clearly indicates that because the author and her husband are retired, they have sufficient time to travel. Answer choices G and H are wordy and are not as clear and effective as answer choice F. Omitting the underlined portion would result in some ambiguity within the sentence: you do not know for what the author and her husband have plenty of time.

5. **The best answer is C.** According to the passage, the fact that the author's grandchildren live in New York combined with the fact that she has two dogs makes it difficult for her to stay for any longer than three weeks in Las Vegas. These responsibilities require her to be at home in New York most of the time. Therefore, it would not be viable for her to stay longer than a few weeks at a time in Las Vegas.

6. **The best answer is J.** The paragraph is written in the present tense, so it is necessary to use the present tense verb *has*. Answer choices F and G use the past tense, so they are incorrect. It is never correct to use the helping verb *has* with the simple past tense verb *became*, so answer choice H is incorrect.

7. **The best answer is C.** If two adjectives modify a noun in the same way, they are called coordinate adjectives. It is necessary to separate coordinate adjectives, such as *warm* and *sunny* with a comma. You could also insert the coordinate conjunction *and*, but you should not use any commas if you do so; therefore, answer choice B is incorrect. A semicolon should be used to separate two related main clauses, so answer choice D is incorrect.

8. **The best answer is G.** The subordinating conjunction *because* clearly indicates that the dogs are the reason that the author cannot fly to Las Vegas. The subordinating conjunction *so* is used to indicate a consequence, and if it is used here, it creates an incomplete sentence; eliminate answer choice F. Answer choice H is a preposition phrase that does not correctly express the idea that having the dogs prevents the author from flying to Las Vegas, so it is incorrect. While answer choice J expresses essentially the same idea as answer choice G, it is wordy and awkward and should be eliminated.

9. **The best answer is A.** It is necessary to use commas to set off parenthetical expressions within a sentence. Because the phrase "both of which are shelter mutts" is not essential to the meaning of the sentence, it is considered a parenthetical, or non-restrictive phrase. A nonrestrictive phrase can be omitted from the sentence without changing the meaning of the sentence.

10. **The best answer is H.** The sentence is written in the present tense, so it is necessary to use the present tense verb *lies*. In addition, the sentence indicates that only one dog *lies* on a blanket on the floor.

11. **The best answer is A.** It is best to leave Sentence 5 where it is in the paragraph. The paragraph describes the behavior of the dogs as they travel with the author, and the sequence of events is clearly indicated without changing the order of the sentences.

12. **The best answer is J.** The passage is written primarily in the present perfect tense so it is necessary to use the present perfect form of the verb *have passed*. The word *past* indicates "the time before the present," and is not a verb that suggests movement. Therefore, eliminate answer choices F and H. Answer choice G is incorrect because it uses the present participle *passing*, which suggests that the action is currently taking place.

13. **The best answer is B.** The context of the sentence indicates that the author paid multiple tolls, so the plural form of the word is correct. The tolls do not possess anything, so it is not necessary to use an apostrophe; eliminate answer choices A and D. Answer choice C includes an extraneous comma, so it is incorrect. Because the phrase "in Oklahoma" is necessary to the meaning of the sentence, it should not be set off by commas.

14. **The best answer is H.** The preceding sentence reads, "I'm mentally preparing myself for a sunless New York sky during our week at home." This sentence and the final sentence of the passage suggest that the author enjoys her visits to "sunny Las Vegas" and is glad for the chance to escape the "sunless New York sky." The sentence reinforces the author's decision to travel back and forth from New York to Nevada.

15. **The best answer is D.** The tone and context of the passage clearly indicate that it is a personal account of the author's travels to and from Nevada. Therefore, it would not fulfill an assignment to write a travel article about Las Vegas, Nevada.

PASSAGE II

16. **The best answer is F.** The indefinite pronoun *one* is used to indicate any of the possible people who acquire a new puppy or dog. Because *one* is a collective pronoun, it is considered a singular pronoun, so the correct verb to use is "acquires." You do not need the pronoun *who* in the sentence, so eliminate answer choice H. It is not correct to use the pronoun *it* refer to people, so eliminate answer choice J.

17. **The best answer is B.** By using the article *the*, the sentence does not clearly indicate to whom the pets belong. The context of the sentence shows that the pets belong to the *dog owners* so it is correct to use the plural pronoun *their*. The word "there" indicates a place, so eliminate answer choice C. The word *they're* is the conjunction of *they* and *are*, so eliminate answer choice D.

18. **The best answer is H.** The sentence is written in the present tense, so it is necessary to use the present tense verb *is*. As it is written, the sentence suggests that the problem will be *become* evident in the future, so eliminate answer choice F. Answer choice G is incorrect because it uses the participle *becoming*; answer choice J is awkward.

19. **The best answer is D.** The sentence discusses the "movements and facial expressions" of "a dog," which means that you should use the singular possessive *dog's*.

20. **The best answer is F.** The sentence is best as it is written. Answer choice G is incorrect because he participle *confusing* suggests that the dog's actions are not clear to the owner, which does not fit the context of the sentence. Answer choice H is incorrect because use of the preposition *by* suggests that the dog finds "waiting for a signal" to be confusing, which is not supported by the context. Answer choice J is incorrect for the same reason; in addition, it is not idiomatic to say "confused with" unless you are making a comparison.

21. **The best answer is B.** Since the sentence refers to a single dog—*the dog*—it is correct to use the singular possessive *its*. As it is written, the sentence is unclear as to whom or what the *lips* belong. It is never correct to use the plural possessive *its*,' because *it* is a singular pronoun. Therefore, answer choice C is incorrect. Answer choice D is incorrect because the sentence refers to a single dog.

22. **The best answer is F.** The paragraph describes several ways that a dog will communicate non-verbally with its owner. The description of a relaxed dog letting "its tongue loll out of its mouth" is another example of this type of communication.

23. **The best answer is A.** This question tests your ability to convey the intended idea clearly and effectively. The simplest way to show that other forms of body language exist is to say "can also indicate." The other answer choices are wordy and awkward and are not appropriate for standard written English.

24. **The best answer is G.** The preceding sentence describes how a dog owner might recognize when his or her dog is being aggressive or fearful. Answer choice G offers a reason for this behavior.

25. **The best answer is A.** The main focus of the paragraph is additional forms of body language that a dog might exhibit when attempting to communicate. Therefore, the best concluding sentence will refer to and generalize the main idea. Only answer

choice A does this. The other answer choices are too specific and offer additional forms of communication.

26. **The best answer is G.** You should use a comma to separate elements that introduce and modify a sentence. The phrase "While most dogs are capable of learning" is an introductory clause that should be followed by a comma. Answer choice F is incorrect because a semicolon should be used to separate two independent clauses. Answer choices H and J are incorrect because it is necessary to use a comma to separate an introductory clause from the rest of the sentence. In addition, answer choice J creates an incomplete sentence.

27. **The best answer is D.** The main focus of the paragraph is on learning to understand a dog's body language and using that knowledge to more effectively train the dog. The sentence that mentions offering a dog treats is least relevant—it does not add any information that is necessary to the flow or structure of the paragraph. While Sentence 6 is not entirely unrelated to the topic, of the answer choices, it is the least relevant to the paragraph.

28. **The best answer is G.** According to the conventions of standard written English, you should never separate a subject from its verb with any sort of punctuation. Therefore, it is incorrect to place a comma, a semicolon, or a colon between *The bottom line*—the subject—and *is*—the verb.

29. **The best answer is A.** The ACT rewards active voice. Therefore, it is better to use the verb *communicating* as opposed to the noun *communication* in this sentence; eliminate answer choices B and D. It is idiomatic to say *involved in*, whereas *involved to* is not correct in standard written English.

30. **The best answer is J.** The ACT rewards active voice. Therefore, it is best to say *to remember*; the other answer choices are not as good because they are written in the passive voice.

PASSAGE III

31. **The best answer is D.** You should use a comma to separate elements that introduce and modify a sentence. The phrase "While the history of the piñata is somewhat murky" is an introductory clause that should be followed by a comma. Answer choice A is incorrect because it creates an incomplete sentence. Answer choice B is incorrect because a semicolon should separate two independent clauses. Answer choice C is incorrect because you must use a comma to separate an introductory clause from the rest of the sentence.

32. **The best answer is F.** The word *where* correctly indicates that most of the piñatas were made in China. Because China is a place, it is better to say *where* than *in*, so answer choice G is incorrect. Answer choice H is incorrect because it includes an extraneous comma. The coordinate conjunction *so* indicates that China is the reason that most piñatas were made to resemble animals, which does not make sense.

33. **The best answer is A.** The underlined portion clearly and effectively indicates that the current custom of filling piñatas with candy or toys differs from the original custom of filling the piñatas with seeds. Answer choice B is incorrect because it includes the ambiguous pronoun *they*. Answer choice C is redundant—it is not necessary to use both *standard* and *customary* in the sentence because the words are synonyms. If you omit the underlined portion, you lose the comparison between the original custom and the current custom.

34. **The best answer is G.** The sentence is written in the past tense so you must use the past tense verb *were kept*. Answer choice H is incorrect because, while it uses the past tense verb *kept*, it creates an incomplete sentence.

35. **The best answer is B.** No commas are necessary to set off a person's name from a clause when the title of occupation precedes the name: for example, President Abraham Lincoln.

36. **The best answer is H.** The paragraph focuses specifically on the piñata and its use in European celebrations. The sentence that the author is considering adding does not offer anything relevant to the main idea of the paragraph.

37. **The best answer is D.** The phrase "that broke easily," while not grammatically incorrect, is redundant. Something that is fragile, by definition, breaks easily. Therefore, answer choices B and C are also redundant, and it would be best to omit the underlined portion.

38. **The best answer is F.** The sentence is describing an event as it might have taken place in the past. Therefore, it is appropriate to use the auxiliary verb *would be* with the past tense verb *hung*. It is not necessary to use any punctuation in the underlined portion.

39. The best answer is B. The past tense verb *dispensed* effectively indicates that the tiny treasures came from the pot after it was broken. The phrase *dispensed with* refers to getting rid of or doing away with something, which does not fit the context.

40. The best answer is H. Only answer choice H uses the active voice to clearly indicate that either the pots could be unadorned, or they could be decorated with ribbons and paper. The other answer choices are in the passive voice and are awkward.

41. The best answer is C. The sentence is discussing more than one piñata, so the plural form *piñatas* is correct. The "s" apostrophe means plural possession, and the apostrophe "s" means singular possession; possession is not indicated by the context of the sentence, so answer choices A and B are incorrect.

42. The best answer is G. The singular subject *every* needs the singular pronoun *its*. The word *it's* is the conjunction of *it is*.

43. The best answer is A. The phrase "In America" is an introductory clause that must be followed by a comma.

44. The best answer is J. The context of the sentence indicates that the piñatas are filled with candy and toys; in fact, the sentence says *pounds*, which can be interpreted as a large quantity. Because *barren* means *empty*, it would not be an appropriate alternative.

45. The best answer is A. As it is written, the underlined portion clearly and effectively shows what the audience most enjoys doing: "watching the blindfolded person swing mightily at nothing but thin air." It is in the active voice and is free from ambiguity. The other answer choices are either in the passive voice or do not make sense.

PASSAGE IV

46. The best answer is J. If a list contains three or more items, it is necessary to separate those items with commas. It is not appropriate to include a comma between the verb *read* and the prepositional phrase *in magazines*, so answer choice F is incorrect. Answer choice G does not include any of the necessary separating commas, so it is incorrect. A semicolon should be used to separate items in a list that follows a colon, so the use of a semicolon in answer choice H is incorrect.

47. The best answer is D. This sentence requires the use of the participle *spending* in order to accurately show that the spending takes place after the watching, reading, and hearing of the advertisements. The helping verb *having* suggests that the spending occurs first, which does not make sense. Answer choice D is the most simple and clear selection.

48. The best answer is F. The underlined portion is best as it is written. It is not appropriate to use a comma to separate a subject from its verb, so answer choice G is incorrect. Both answer choices H and J create an incomplete sentence that lacks a main verb.

49. The best answer is A. The underlined portion correctly uses the contraction of *you are*, whereas answer choice B incorrectly uses the possessive pronoun *your*. The other answer choices are awkward and contain extraneous commas.

50. The best answer is J. The underlined portion is redundant, because the sentence already indicates that the person is *special*. Therefore, the best thing to do is to omit the underlined portion. Because all of the answer choices are grammatically correct and express the same idea, you should be able to quickly recognize the redundancy.

51. The best answer is B. This question tests your ability to recognize idiom. In standard written English it is appropriate to use the phrase *filled with*.

52. The best answer is H. Because the sentence uses the pronoun *you* earlier in the sentence, it is not necessary to restate the pronoun later in the sentence. It is clear that *you* are performing both actions—decorating a picture frame and using nuts and bolts.

53. The best answer is D. The coordinate conjunction *and* is not necessary at this point in the sentence. The sentence presents a list of items that could be used in a shadow box, and *a dried flower* is an item that falls in the middle of the list. Therefore, no additional language is necessary.

54. The best answer is F. In order to avoid redundancy and wordiness, it is best to simply say *gather from*. This clearly and effectively expresses the idea.

55. The best answer is D. Sentence 1 is an appropriate introductory sentence because it includes the transitional phrase *For example*, and then goes on to give examples of how to create a handmade gift from someone's trinkets or other personal items. Sentence 2 directly follows with yet another example. Sentence 4 would best follow sentence 2,

because it provides details on how to make a shadow box. Finally, sentence 3 is best as the last sentence because it effectively concludes the topic discussed in the paragraph.

56. The best answer is G. The sentence as it is written is an incomplete sentence that lacks a main verb. Answer choice G is correct because it is necessary to use a comma to separate an introductory clause from the rest of the sentence. Answer choice H has the comma after the verb, which is incorrect.

57. The best answer is D. It is best to use the adverb *nearly* to modify the adjective *endless*. Answer choice D is best because it does not include the extraneous verb *to be*.

58. The best answer is H. To identify more effectively exactly what is happening, it is better to use the specific pronoun *this*. The pronoun *it* is rather ambiguous, and the coordinate conjunction *so* is not appropriate.

59. The best answer is A. The student benefits by learning about scrapbooking and the store benefits be receiving customer patronage. The other answer choices offer support for why a patron would take a class, but not for why a store owner would offer such a class.

60. The best answer is J. The notion that original, hand-made pictures or books will be enjoyed is stated previously in the paragraph. Therefore, it is not necessary or relevant to mention it again. The other answer choices are not supported by the essay.

PASSAGE V

61. The best answer is D. No punctuation is necessary in this sentence. A good rule of thumb is to use a comma wherever you would naturally pause in a sentence. Because there are no natural pauses in this sentence, you don't need a comma. Answer choice B is incorrect because a semicolon should be used to separate two independent clauses. Answer choice C is incorrect because a colon should generally be used to introduce a list.

62. The best answer is J. Answer choice J is best because it indicates not only that accidents occur more frequently, but how many times more frequently they occur. This is more detailed and relevant than the information contained in the other answer choices.

63. The best answer is B. The plural subject *These statistics* requires the plural verb *take*. The other answer choices use the wrong verb form or tense.

64. The best answer is F. The sentence in answer choice F introduces the idea of *ingenious designs* which are then discussed in the paragraph. Answer choice G does not provide an effective transition, and answer choice H is irrelevant to the topic of the paragraph. Answer choice J is incorrect because the focus of the passage is safety.

65. The best answer is C. In order to maintain parallelism within the sentence, it is necessary to use *illuminating* to match the verb *allowing*.

66. The best answer is G. The plural noun *headlights* requires the plural verb *are*.

67. The best answer is B. You should use a semicolon to separate two independent but related clauses. A comma is insufficient and creates a comma splice, so answer choice A is incorrect. Answer choices C in incorrect because it does not include any punctuation. Answer choice D is incorrect because it is not correctly punctuated.

68. The best answer is H. By using the verb phrase *doing so*, answer choice H clearly and effectively indicates that using the high-beams would provide more light. Answer choice F is incorrect because the pronoun *they* is ambiguous—it is unclear whether the antecedent is *high-beams* or *drivers*. Answer choice G is incorrect for a similar reason—the pronoun *it* is ambiguous. Answer choice J is awkward and contains an ambiguous pronoun.

69. The best answer is D. It is best to say *switching proves* because *switching* is singular. The other answer choices include incorrect verb tense and aspect.

70. The best answer is G. It is appropriate to use a colon to introduce a list of items. Answer choice F is incorrect because it creates an incomplete sentence. Answer choice H is incorrect because it includes a comma splice. Answer choice J is incorrect because it is awkward and wordy.

71. The best answer is B. To clearly indicate that the images are displayed on a screen after the images are detected, it is best to use the phrase *are then displayed*. The images themselves do not display; rather, they are displayed, so answer choices A and C are incorrect.

72. The best answer is G. Sentence 5 should be the last sentence of the paragraph because it effectively

concludes the topic of the paragraph, and most logically follows sentence 6.

73. The best answer is D. The phrase "particularly during dangerous nighttime driving" is a nonrestrictive clause, which should be set off with commas. Only answer choice D correctly places a comma between *road* and *particularly*.

74. The best answer is F. Use a comma with a coordinating conjunction to separate main clauses within a sentence. A coordinating conjunction connects words, phrases, or clauses that are of equal importance in the sentence.

75. The best answer is B. Paragraph 1 states that more accidents occur at night than in the daytime. The new sentence uses the possessive determiner *these*, which takes the mention of night accidents as its antecedent from earlier in the paragraph.

Mathematics Test Explanations

1. **The correct answer is A.** You are given that one foot is approximately 0.3048 meters. Therefore, a building that is 65 feet long will be 65(0.3048), or 19.8 meters long.

2. **The correct answer is H.** To calculate the new hourly rate, multiply the current rate by 18%, or 0.18, its decimal equivalent, and add the result to the current rate:

$$30(0.18) + 30 = 30 + 5.40, \text{ or } \$35.40$$

3. **The correct answer is D.** To find the average, divide the sum of the contributions by the number of contributions, as follows:

$$\frac{(25 + 40 + 30 + 15)}{4} = \frac{110}{4} = 27.50$$

4. **The correct answer is F.** To find the difference in the distance traveled, first find the distance traveled by each bus. Distance equals rate multiplied by time. Since Bus X travels 40 miles per hour for 2 hours, it traveled $40(2) = 80$ miles. Likewise, since Bus Y travels 60 miles per hour for $1\frac{1}{2}$ hours, it traveled $60(1.5) = 90$ miles. Therefore, the difference is $90 - 80$, or 10 miles.

5. **The correct answer is C.** The expression $(r + 2)(r - 3)$ will equal zero when either $r + 2$ or $r - 3$ equals zero. Thus $r = -2$ or $r = 3$.

6. **The correct answer is H.** The perimeter is the distance around the parallelogram. In parallelograms, opposing sides have equal lengths. If PS is 7 cm long, so is QR. Also, PQ and SR will have the same length. Set the length of PQ and SR to l, and solve.

$$7 + 7 + l + l = 40$$

$$14 + 2l = 40$$

$$14 + 2l = 40$$

$$l = \frac{26}{2}, \text{ or } 13.$$

7. **The correct answer is D.** If the x-coordinate of each point on a line is 5 more than half the y-coordinate, then $x = \frac{y}{2} + 5$. To find the slope of the line, solve for y and put the equation in slope-intercept form ($y = mx + b$, where m is the slope). To do so, first subtract 5 from both sides, then multiply the entire equation by 2 to get $y = 2x - 10$. The slope is 2.

8. **The correct answer is J.** You are given that a rectangular garden has a length of x and a width of y, and has its length reduced by 3 feet and its width extended by 2 feet. Therefore, the length becomes $x - 3$ and its width becomes $y + 2$. The area of the new garden is then $(x - 3)(y + 2)$.

9. **The correct answer is A.** To get y in terms of x and z, solve $x = 3yz^2$ for y, as shown below:

$$x = 3yz^2$$

$$\frac{x}{3z^2} = y, \text{ or } y = \frac{x}{3z^2}$$

10. **The correct answer is H.** The triangle pictured is isosceles, meaning that the two angles opposite the sides that have equal length have equal measure. Since the sum of the interior angles of a triangle is 180°, the measure of $\angle\alpha$ is $180° - 2(55°)$, or $180° - 110°$, which is 70°.

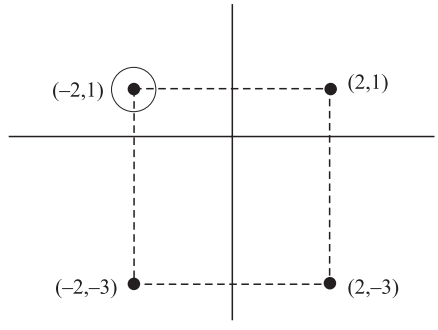
11. **The correct answer is C.** To solve this problem, distribute using the FOIL method, as follows:

$$(3x^2 - 1)(x^2 - 4)$$

First: $3x^4$
Outside: $-12x^2$
Inside: $-x^2$
Last: 4

Combine like terms and simplify to get $3x^4 - 13x^2 + 4$.

12. **The correct answer is K.** To solve this problem it might be helpful to draw a picture like the one below.



Since the figure is a square, the placement of the missing point becomes clear: the upper left corner. The point will share the same x-coordinate as $(-2,-2)$ and the same y-coordinate as $(2,1)$, making its coordinates $(-2,1)$.

13. The correct answer is B. To reduce the expression, recall that when two numbers that consist of the same base raised to a power are divided, you should subtract the exponents. Thus $\frac{x^8}{x^4} = x^{8-4} = x^4$. Likewise, the expression $\frac{(x^8y^{12})}{(x^4y^3z^2)} = \frac{(x^{8-4}y^{12-3})}{z^2}$, which is equivalent to $\frac{x^4y^9}{z^2}$.

14. The correct answer is H. Logarithms are used to indicate exponents of certain numbers called bases. By definition, $\log_a b = c$, if $a^c = b$. Therefore, $\log_n 64 = 2$ implies that $64 = n^2$. Thus $n = 8$.

15. The correct answer is B. To find the percentage of the 700 high-school students who answered that the Warriors were their favorite team, divide the number of students who answered Warriors by 700 and multiply by 100%. To find the number of students who answered Warriors, add the total number of students who answered that they liked a different team and subtract that quantity from 700. Thus the percentage of students who answered Warriors is $\frac{(700-(250+200+50))}{700} \times 100\%$, or $\frac{(700-500)}{700} \times 100\%$, which is $\frac{200}{700} \times 100\%$, or approximately 28.6%.

16. The correct answer is H. If $x^2 = 36$ and $y^2 = 81$, then $x = \pm 6$ and $y = \pm 9$. There are then 4 possible combinations of $x + y$: $6 + 9$, $-6 + 9$, $6 + (-9)$, and $-6 + (-9)$, which have values 15, 3, -3, and -15, respectively. Therefore, the value of $x + y$ cannot be 0.

17. The correct answer is B. To see the properties of each linear equation more clearly, convert each to slope-intercept form ($y = mx + b$, where m is the slope and b is the y-intercept). The equation $4y = 3x + 12$ becomes $y = \frac{3x}{4} + 3$ after you divide by 4. The equation $-4y = -3x - 8$ becomes $y = \frac{3x}{4} + 2$ after you divide by -4. It is now clear that the equations are lines with slope $\frac{3}{4}$, making them parallel lines with positive slope. However, since they have different y-intercepts, they are two distinct lines.

18. The correct answer is G. The absolute value of any number is non-negative. Since $|-3| = 3$, the value of $\frac{-6}{|-3|} = \frac{-6}{3}$, or -2.

19. The correct answer is A. In order for the expression $(a + y)(a + z)$ to equal zero, either $a + y = 0$ or $a + z = 0$. Thus either $a = -y$ or $a = -z$.

20. The correct answer is K. The statement that CF is congruent to EF is not true because ΔCEF is an isosceles triangle in which CE and CF are equal. Since the triangle is isosceles, $\angle CFE$ is congruent to $\angle CEF$, and they have measure 70°. In a triangle, the length of a side is proportional to the measure of the angle opposite it. Since the measure of the angle opposite EF is only 40° while the measure of the angle opposite CF is 70°, EF must be shorter than CF.

21. The correct answer is E. To find the slope, convert the equation $21x - 3y + 18 = 0$ to slope-intercept form ($y = mx + b$, where m is the slope and b is the y-intercept). To do so, first subtract 18 and $21x$ from both sides to get $-3y = -21x - 18$. Then divide both sides by -3 to get $y = 7x = 6$. Therefore the slope is 7.

22. The correct answer is J. The least common denominator (LCD) is actually the least common multiple of the three denominators. To find the LCD, first find the greatest common factor of the denominators. In this case, each denominator contains either a, b, or both b and c, so those are the factors that they have in common. The values that remain in the three denominators represent what they don't have in common, and are the least common multiples of each denominator. To find the least common multiple, otherwise known as the least common denominator (LCD), simply multiply the least common multiples of each denominator together, to get $a^2 \times b^2 \times c^2$.

23. The correct answer is C. To solve this problem, go through and try each answer choice to see which one has the desired result. The correct answer is -2 because adding negative 2 is the same as subtracting 2: $\frac{(5-2)}{(8-2)} = \frac{3}{6}$, or $\frac{1}{2}$.

24. The correct answer is G. To solve this problem, first solve the equation $2y = 16$ for y to get $y = 8$. Then substitute $y = 8$ into the equation $x + y = 13$ to get $x + 8 = 13$, making $x = 5$.

25. The correct answer is B. One possible method of solving this problem is to systematically eliminate wrong answer choices. It is given that $|m| > |n|$.

Assuming that statement is true, then m cannot equal n because if $m = n$, then $|m| = |n|$; eliminate answer choice A. Pick numbers for the variables to more clearly see the relationships, as follows:

Answer choice B: When $m = 3$ and $n = 2$, $|m| > |n|$ is true. Likewise, when $m = -3$ and $n = -2$, $|m| > |n|$ is true. Therefore, answer choice B is correct.
Answer choice C: When $m = 2$ and $n = 3$, $|m| > |n|$ is not true, so answer choice C is incorrect.
Answer choice D: When $m = 3$ and $n = -4$, $|m| > |n|$ is not true, so answer choice D is incorrect.
Answer choice E: Because you are not given any information about n, you cannot determine a relationship, so answer choice E is incorrect.

26. The correct answer is G. A graph crosses the x-axis at the point when $y = 0$. Given that $y - 5 = \frac{1}{2x} + 1$, let $y = 0$ such that $-5 = \frac{1}{2x} + 1$. Subtracting 1 from both sides yields $-6 = \frac{1}{2x}$. Multiplying by 2 yields $-12 = x$.

27. The correct answer is D. To calculate the area of the living room, first calculate its dimensions. The length of the living room is 18′ less the width of the hallway, which is 4′, making it $18 - 4 = 14'$. The width of the living room is 20′ less the width of the kitchen, which is 8′, making it $20 - 8 = 12'$. Thus the area of the living room is $14' \times 12' = 168$ square feet.

28. The correct answer is F. The perimeter is equal to the distance around an object. To calculate the perimeter of the ground floor of the townhouse, add the sides. The perimeter of the ground floor of the townhouse is $2(20') + 2(18')$, or $40' + 36'$, which is 76′.

29. The correct answer is B. This question asks you to calculate the percentage by which the population has decreased. First, determine the number of birds that you would expect to find in your sample area if the population density had not changed. Multiply the population density 3 years ago (20 birds) by the size of the sample area (3.25 acres):

$$20 \times 3.25 = 65$$

You would expect to find 65 birds in your sample area if there had been no change in population density.

Next, subtract the actual number of birds recorded in the sample area (47) from the expected number of birds (65):

$$65 - 47 = 18$$

There are 18 fewer birds in the sample area than what you would expect to find. Since you are asked to calculate the percentage decrease, set up a proportion:

18 birds is to 65 birds as x percent is to 100 percent.

$$\frac{18}{65} = \frac{x}{100}.$$

Cross-multiply and solve for x:

$$65x = 1{,}800$$
$$x = \frac{1{,}800}{65}$$
$$x = 27.69$$

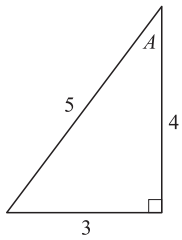
Since the problem asks for an answer to the nearest tenth, round to 27.7. According to the study, the population density of the particular bird that you are studying has decreased by approximately 27.7%.

Answer choice C could be obtained if you found the difference between the number of birds expected (65) and the number of birds recorded (47), then divided by the number of birds recorded (47) to get a percent. Answer choice D is the percentage that the number of birds per acre 3 years ago (20) is of the number of birds recorded in 3.5 acres this year (47).

30. The correct answer is H. To find the product $(\tan\alpha)(\sin\beta)$, break up $\tan\alpha$ and $\sin\beta$ to their ratios. The ratio $\tan\alpha$ is the side opposite α divided by the side adjacent to α: $\frac{z}{x}$. The ratio $\sin\beta$ is the side opposite β divided by the hypotenuse: $\frac{x}{y}$. The product $(\tan\alpha)(\sin\beta)$ is, therefore, $\left(\frac{z}{x}\right)\left(\frac{x}{y}\right) = \frac{z}{y}$.

31. The correct answer is C. In order to add $\frac{1}{x} + \frac{3}{4}$, the fractions must have a common denominator. To achieve a common denominator, multiply $\frac{1}{x}$ by $\frac{4}{4}$ and $\frac{3}{4}$ by $\frac{x}{x}$ to get $\frac{4}{4x} + \frac{3x}{4x}$, which equals $\frac{(4 + 3x)}{4x}$.

32. The correct answer is F. To solve this problem, it may be helpful to draw a picture in which one angle of a right triangle is labeled A, as shown below:



Because $\cos A = \frac{4}{5}$, let the side adjacent to angle A be 4 and the hypotenuse be 5. Likewise, since $\sin A = \frac{3}{5}$, let the side opposite angle A be 3 and the hypotenuse be 5. It follows then that $\tan A =$ side opposite divided by side adjacent, or $\frac{3}{4}$.

33. The correct answer is A. To solve this problem, convert $5x + 3y = 8$ to the slope-intercept form, $y = mx + b$, where m is the slope and b is the y-intercept. To do so, first subtract $5x$ from both sides to get $3y = -5x + 8$. Dividing the entire equation by 3 yields $y = -\frac{5x}{3} + \frac{8}{3}$. Thus $\frac{8}{3}$ is the y-intercept.

34. The correct answer is J. To divide $\frac{a^x}{a^y}$, subtract the exponents. Thus $\frac{a^x}{a^y} = a^{x-y}$. If $\frac{a^x}{a^y} = a^4$, then $a^{x-y} = a^4$, making $x - y = 4$.

35. The correct answer is C. Between May and August, there were 3 price decreases (in June, July, and August). If the price was decreased by 20%, then the resulting price was 80% of the previous month's price. Thus, in June the price was $0.8d$; in July the price was $0.8(0.8d)$; in August the price was $0.8(0.8(0.8d))$, which is equivalent to $(0.8)^3d$, or $0.512d$.

36. The correct answer is K. Given that $|5 - 2x| > 5$, then either $5 - 2x > 5$ or $5 - 2x < -5$. In the case that $5 - 2x > 5$, $-2x > 0$ making $x < 0$ (when you divide by a negative number remember to switch the direction of the inequality). In the case that $5 - 2x < -5$, then $-2x < -10$ making $x > 5$. Thus the range for x is $x < 0$ or $x > 5$. Of the answer choices, only 6 fits into the range for x.

37. The correct answer is E. To solve this problem, first distribute then combine like terms. Distribute 0.1 as follows:

$$0.1(t + 3{,}420) = t$$
$$0.1t + 342 = t.$$
$$342 = 0.9t.$$
$$t = \frac{342}{0.9} = 380$$

38. The correct answer is J. Any line parallel to the y-axis is a vertical line. Vertical lines have slopes that are undefined. Remember that the definition of slope is rise/run; vertical lines have no run and thus dividing rise by run is dividing by 0, making the quotient undefined.

39. The correct answer is E. To find which equation has the smallest slope, first convert any equations to slope intercept form ($y = mx + b$, where m is the slope and b is the y-intercept) if they are not already in that form. In choice E, dividing both sides of $7y = 3x - 7$ by 7 yields $y = \frac{3x}{7} - 1$, which has a slope of $\frac{3}{7}$. When compared to the other slopes, $\frac{3}{7}$ is the smallest (the next closest is $\frac{1}{2}$, which is only slightly larger than $\frac{3}{7}$).

40. The correct answer is K. Let the number of minutes it would take Amy to run 10.3 miles be y. If she ran at the same pace for 10.5 mile as she did for 3.5 miles, then the rates in miles per minute would be equal, making $\frac{3.5}{x} = \frac{10.5}{y}$. Since the question asks for the number of minutes it would take Amy to run 10.5 miles, solve $\frac{3.5}{x} = \frac{10.5}{y}$ for y. To do so, first invert both sides of the equation to get $\frac{x}{3.5} = \frac{y}{10.5}$. Then multiply both sides by 10.5 to get $y = \frac{10.5x}{3.5}$, or $3x$.

41. The correct answer is A. Regardless of the dimensions of a rectangle, tripling the length and width will always have the effect of tripling the perimeter because perimeter is directly proportional to length and width ($P = 2l + 2w$).

42. The correct answer is H. To solve this problem, multiply the expression $(x + 3)$ by $(x + s)$ to get $x^2 + 3x + sx + 3s$. You are given that $x^2 + rx + 12$ is equivalent to $x^2 + 3x + sx + 3s$.

Therefore, $3s$ is equal to 12, making s equal to 4. It is also apparent that $3x + sx$ is equivalent to rx.

Set the quantities equal and solve for r, as follows:

$$rx = 3x + sx$$
$$rx = x(3 + s)$$
$$r = 3 + s$$

Because $s = 4$, r must equal 7.

43. The correct answer is C. Systems of equations have an infinite number of solutions when the equations are equivalent (*i.e.* they graph the same lane). In order for the two equations to be equivalent, the constants and coefficients must be proportional. If the entire equation $3x + 5y = 27$ is multiplied by 4, the result is $4(3x + 5y) = 4(27)$, or $12x + 20y = 108$. Thus, in order for the two equations to be equivalent, $3b = 108$, or $b = 36$.

44. The correct answer is J. Any integer when multiplied by an even integer results in an even integer. Also, the addition of any two even integers yields another even integer. Thus, regardless of the value of a or the power of a, multiplying by an even integer and adding an even integer will yield an even integer. This only occurs in the calculation $6a^4 + 6$.

45. The correct answer is B. To solve this problem, remember that $|3a - 3| \geq 12$ is equivalent to $3a - 3 \geq 12$ or $3a - 3 \leq -12$. Adding 3 to both sides and dividing by 3 yields $a \geq 5$ or $a \leq -3$.

46. The correct answer is H. To find $\cos\frac{5\pi}{12}$ using $\cos(\alpha + \beta) = (\cos\alpha)(\cos\beta) - (\sin\alpha)(\sin\beta)$ given that $\frac{5\pi}{12} = \frac{\pi}{4} + \frac{\pi}{6}$, you can first substitute $\frac{\pi}{4}$ for α and $\frac{\pi}{6}$ for β and get $\cos\left(\frac{\pi}{4} + \frac{\pi}{6}\right) = \left(\cos\frac{\pi}{4}\right)\left(\cos\frac{\pi}{6}\right) - \left(\sin\frac{\pi}{4}\right)\left(\sin\frac{\pi}{6}\right)$. Using the table of values to substitute into that equation, you get $\cos\frac{5\pi}{12} = \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$, or $\frac{(\sqrt{6} - \sqrt{2})}{4}$.

47. The correct answer is A. To solve this problem, factor out the x in the numerator of the fraction and the 3 in the denominator to get $\frac{(xy - xz)}{(3y - 3z)} = \frac{x(y - z)}{3(y - z)}$. Since the quantity $y - z$ is on the top and bottom, it cancels out, leaving $\frac{x}{3}$. Since $\frac{(xy - xz)}{(3y - 3z)} < 0$, it follows that $\frac{x}{3} < 0$. Multiplying both sides of the inequality by 3 yields $x < 0$. Thus x can be any number less than 0 (all negative numbers).

48. The correct answer is G. If the perimeter of a square is 36 units, then each side is 9 (since perimeter in a square is $4(s)$ where s is the length of a side). To find the length of the diagonal, you can use the Pythagorean Theorem because the diagonal is the hypotenuse of a right triangle with legs of length 9. Thus $d^2 = 9^2 + 9^2$, or $81 + 81$, which equals 2(81), and $d = \sqrt{(2(81))}$, or $\sqrt{2}\sqrt{81}$, which is $9\sqrt{2}$.

49. The correct answer is E. A circle with center (h, k) and radius r has equation $(x-h)^2+(y-k)^2 = r^2$. In order to determine the equation of the circle in the standard (x, y) coordinate plane that has a radius of 4 units and the same center as the circle determined by $x^2 + y^2 - 6y + 4 = 0$, first determine the center of the circle defined by $x^2 + y^2 - 6y + 4 = 0$ by converting the equation to the standard form for a circle (complete the square):

$$x^2 + y^2 - 6y + 4 = 0$$
$$x^2 + y^2 - 6y + 9 + 4 - 9 = 0$$
(no net change in value)
$$x^2 + (y^2 - 6y + 9) + 4 - 9 = 0 \quad \text{(regroup)}$$
$$x^2 + (y - 3)^2 + -5 = 0 \quad \text{(factoring)}$$
$$x^2 + (y - 3)^2 = 5$$

Thus the center of the circle is (0,3). A circle that has a radius of 4 units and the same center has the equation $x^2 + (y - 3)^2 = 16$.

50. The correct answer is H. A rectangular kitchen that is 8 feet longer than it is wide with width w would have length $l = w + 8$, or $w = l - 8$. Since the area $A = 240 = lw$, substitute $w = l - 8$ into $240 = lw$ to get $240 = l(l - 8) = l^2 - 8l$, or $l^2 - 8l - 240 = 0$. To find l, factor $l^2 - 8l - 240 = 0$ into $(l + 12)(l - 20) = 0$. Thus $l = -12$ (which doesn't make sense because a kitchen cannot have a negative length), or $l = 20$.

51. The correct answer is C. Parallel lines have equal slopes. To find the slope of a line that is parallel to the line determined by the equation $5x - 4y = 8$, put the equation in slope-intercept form ($y = mx + b$). To do so, first subtract $5x$ from both sides to get $-4y = -5x + 8$. Then divide by -4 to get $y = \frac{5x}{4} - 2$. Thus the slope is $\frac{5}{4}$.

52. The correct answer is H. To solve this problem, recall that $81 = 3^4$. Then $81^{3x-2} = (3^4)^{3x-2} = 3^{4(3x-2)}$. Further, if $3^{8x} = 81^{3x-2}$, then $3^{8x} = 3^{4(3x-2)}$ and $8x = 4(3x-2)$, $12x - 8$. Subtracting $8x$ from both sides and adding 8 to both sides yields $4x = 8$, or $x = 2$.

53. The correct answer is B. Since the width of the frame is $\frac{5}{8}$ or 0.625 inches, the length of the viewable portion is $30 - 2(0.625) = 28.75$ inches and the width is $18 - 2(0.625) = 16.75$ inches. Thus, the area is $28.75 \times 16.75 = 481.56$ square inches.

54. The correct answer is F. If a horse eats 12 bales of hay in 5 days, the average rate is $\frac{12}{5}$ bales per day. At this rate, the number of bales of hay that the horse eats in $5 + x$ days is the 12 bales for the 5 days plus $\frac{12}{5}$ bales per day after that, or $\left(\frac{12}{5}\right)x$. Thus the total is $12 + \left(\frac{12}{5}\right)x$, or $12 + \frac{(12x)}{5}$.

55. The correct answer is A. To solve this problem, substitute the equation $x = -5$ into $y = x - 5$ to find the y-coordinate at which the lines $x = -5$ and $y = x - 5$ intersect (the x-coordinate is -5 because it is given that $x = -5$). Thus $y = (-5) - 5 = -10$. The point of intersection is $(-5,-10)$.

56. The correct answer is J. The number of miles a runner must travel in a 4-lap race where the course is a circle of radius m miles will be equal to 4 times the circumference of the circle. Since circumference is $2\pi r$, where r is the radius, the circumference is $2\pi m$. Since it is a 4-lap race, the total number of miles traveled is $4(2\pi m)$, or $8\pi m$.

57. The correct answer is B. Given that the graph of the line $y = (n + 1)x + 6$ in the standard (x,y) coordinate plane passes through (4,8), plug in the values of the point (4,8) into the equation and solve for n. Substituting (4,8) into $y = (n+1)x+6$ yields $8 = (n+1)(4)+6 = 4n+10$. To solve $8 = 4n+10$, subtract 10 from both sides and divide by 4 to get $n = -\frac{1}{2}$.

58. The correct answer is J. You are given that a computer repair person charges \$50.00 per hour, plus an additional mileage fee which varies directly with the square root of the number of miles traveled. Therefore, the total fee can be expressed as $50h + k\sqrt{m}$, where h is the number of hours worked, m is the number of miles traveled, and k is some constant. Since one hour plus 25 miles traveled costs $\$140, 140 = 50 + k\sqrt{25} = 50 + 5k$. Since $140 = 50 + 5k$, $k = 18$. The total amount charged for one hour plus 36 miles traveled is $50+18\sqrt{36} = 50 + 18(6) = 50 + 108 = \$158.00$.

59. The correct answer is C. Since $\sin Z$ is the ratio of the side opposite the angle to the hypotenuse, the side opposite Z, which is XY, must be determined. To do so, apply the Pythagorean Theorem, as follows:

$$10^2 = XY^2 + 4^2$$
$$XY^2 = 100 - 16 = 84$$
$$XY = \sqrt{84} = \sqrt{((21)(4))} = \sqrt{21}\sqrt{4}, \text{ or } 2\sqrt{21}.$$

Since $\sin Z$ is the ratio of the side opposite the angle to the hypotenuse, $\sin Z = \frac{2\sqrt{21}}{10}$

60. The correct answer is J. The image of a point (x, y) reflected across the line $y = x$ will have coordinates (y, x). If the coordinates of point A are (m, n), then the coordinates of point A' are (n, m).

Reading Test Explanations

PASSAGE I

1. **The best answer is C.** The passage provides several interpersonal and physical examples of teammates working together for their goal of winning the game. Mark's quotation is an example of encouragement. The narrator describes the role of linemen in pushing back the defense for the running back. The other answer choices are not supported by the passage.

2. **The best answer is J.** The passage gives details about the almost meditative quality of Joe's preparation for the snap of the football. According to the passage, "instinct told him what he needed to know." *Engrossed* is the quality of being deeply involved in something. The narrator uses rich descriptions of the rituals of the football play to emphasize the players' dedication to the game.

3. **The best answer is B.** The passage discusses Joe's surprise at the final outcome of the game. He first saw what had happened, before it truly sank in that the game had been lost. This correlates with answer choice A. The passage then goes on to state that "'We had them beat, you know,' Mark hissed to Joe as they walked slowly off the field." This illustrates both commiseration with Mark, and that he may have been "trudging" off the field in walking slowly with his teammates. The only answer choice not supported by details in the passage is answer choice B. Nowhere in the passage does it state that Joe attacked his opponents following the game.

4. **The best answer is G.** The first paragraph provides details on the setting that reinforce the intensity of the game so far. "A cold wind soothed the faces" indicates the faces are hurt or uncomfortable. The third sentence describes breathing dirty air. The word "muster" implies considerable effort on the part of the lungs. The phrases "game-ending" and "season-defining" indicate the team is at a crucial moment in the game.

5. **The best answer is D.** The last sentence of the passage describes how the loss is particularly devastating because of how close Joe's team came to winning on the final play. The other choices are not supported by the passage.

6. **The best answer is G.** Details in paragraphs 4 and 5 describe Joe's careful analysis of the defense made prior to the start of the play. The sentence "Joe glared at his opponents, pleased . . ." is the best example to support answer choice G. The next sentence provides further evidence.

7. **The best answer is D.** Mark's quotation from the last paragraph shows that the other team was more fatigued at the end of the game than was his team. The sentence "'We should have won the game'" shows that Mark believed the team was prepared to win but failed.

8. **The best answer is H.** The last sentence of paragraph 3 describes the hand clasping, which immediately follows Mark's statement of encouragement to his teammates. These two events come at the end of the huddle before the play. The purpose of the huddle is to establish the play and, hence, the upcoming roles of the individual players. The hand-clasping ritual seems to tie the team back together momentarily before they must break apart to perform their individual tasks during the play.

9. **The best answer is A.** The passage describes Joe being pleased by the block and "assuming" it had sent his man back "more than enough to free the rusher to enter the endzone." This best supports answer choice A.

10. **The best answer is H.** Mark's quotation in paragraph 3 is an example of the magnitude of the game (and so, the final play) in the context of a season that includes "practicing for four months." Mark even alludes to the success of the final play reflecting the quality of the team developed over four years.

PASSAGE II

11. **The best answer is B.** The first paragraph states that the tourist was "looking for a souvenir that represents the local culture." However, when he stopped to browse at a small street vendor, he was shocked to see a plethora of American items. This best supports answer choice B. The other answer choices are not supported by the context of the passage.

12. **The best answer is H.** As written in the passage, "American logos appear in quite unexpected places, embodying the *ubiquitous* American symbols worldwide." Because American symbols appear in unexpected places worldwide, it makes the most sense that these symbols would be described as "far-reaching." Answer choice F is incorrect; the price of these symbols is not mentioned anywhere in the passage. Answer choice G does not make sense in the context of the passage. Answer choice J does not make sense because if American symbols were localized, they would be found solely in America.

13. **The best answer is D.** It is stated in the passage that those people who reject Americana do so because they "lament the steady decline of distinct national, ethnic, and cultural identities as omnipresent American influences overpower ancient traditions and beliefs."

14. **The best answer is H.** The passage explicitly states that jobs have been created due to the spread of American influence, and that with American commerce comes the modernization of infrastructure. The passage does not state that declines in tourism result from the spread of American influences, but rather that American businesses are quickly spreading to tourist areas abroad.

15. **The best answer is B.** When the passage states that "The local cantinas and native boutiques are losing the battle against large American corporations," it is clear that these places are straining to even stay in business and retain their cultural uniqueness versus the American businesses that are invading the resort towns. The other answer choices are not supported by details in the passage.

16. **The best answer is G.** The passage states that Japan is a "prime example of where American involvement has been beneficial," and that "America left a pervasive cultural footprint on the country" while maintaining "a distinct culture that is rich in the traditions of the past." This clearly indicates that American influence was beneficial to Japan and that a well-balanced mix of American and Japanese culture ensued. This best supports answer choice G. The other answer choices are beyond the scope of the passage.

17. **The best answer is C.** The passage illustrates both a country that has been helped by American influence (Japan) and a country whose native businesses are suffering due to American influence (Mexico). It is unclear what will happen if American cultural influences continue to penetrate foreign markets. Answer choice B may appear to be correct; however, the example of America's influence on Japan negates this option.

18. **The best answer is J.** Paragraph 3 discusses those people who are rejecting "American 'cultural imperialism,'" and states that these people "*lament* the steady decline of distinct national, ethnic, and cultural identities as omnipresent American influences overpower ancient traditions and beliefs." Because these people are rejecting this spread of American influence, it does not make sense that they would embrace or enjoy the loss of other cultures. It also does not make sense that they would deny the steady decline of distinct national cultures, as it is something that is obviously occurring. To "regret" is to express grief over or be unhappy with. The passage illustrates that the people who are rejecting American cultural imperialism are unhappy with its consequences.

19. **The best answer is A.** As stated in the passage, "What he sees shocks him: T-shirts and posters promoting American football, basketball, and baseball teams." This clearly illustrates the fact that the street vendor was selling American sports memorabilia. The other answer choices are not supported by details in the passage.

20. **The best answer is F.** The last paragraph of the passage states that "Many benefits and problems are created by American commercial and cultural expansion abroad." While there are some drawbacks to American expansion, there are still gains. This best supports answer choice F. Answer choice H may appear to be correct; however, the last paragraph does not clarify whether problems from American expansion in the future will be greater than benefits.

PASSAGE III

21. **The best answer is B.** The passage asks "how a person is able to paint or draw something so lifelike and *emotive*," and marvels at "what stirred these men to put the brush to the canvas so delicately." "Emotive" means "causing strong feelings," which is also a synonym for "stirring." This is the most logical choice based on the context of the passage.

22. **The best answer is H.** Realism is noted in the passage as one of the first formal styles of art, preceding Impressionism. The passage later states that among the styles of painting evolving from Impressionism were "Cubism, Abstract Art, Expressionism, Abstract Expressionism, Modernism, and a host of other styles . . ." Realism is the only style of art not mentioned in this list.

23. **The best answer is B.** As stated in the passage, Realist painters "focused on depicting real life unembellished with fanciful notions or feelings." This correlates with answer choice B. Answer choice C may appear to be correct because the passage makes note of unattractive physical attributes. However, the passage simply states that artists simply did not "gloss over" unattractive physical attributes of their subjects; artists portrayed their subjects as they actually looked. Unattractive features were not the focus of Realist artists.

24. The best answer is G. The passage clearly states that though Realism fell out of favor for a while, it was "revived during the Renaissance," and remained popular for several centuries thereafter. This best supports answer choice G. The other answer choices are not supported by details found in the second paragraph.

25. The best answer is B. The passage begins with one of the first popular formal styles of art. The passage then goes on to describe different styles of art that have evolved over time. This best supports answer choice B. Answer choice A is incorrect because the passage does not discuss the changing world and how painters related to it. Answer choice C is incorrect because the passage only mentions the popularity of Realism falling and rising again; the passage does not discuss the popularity of other styles of painting. Answer choice D is incorrect because the passage does not discuss artists adapting to popular demand.

26. The best answer is J. The passage states that while some styles of art that grew out of Impressionism became their own schools of thought, others were simply an "artist's trademark way of painting." In this context, speaking of styles of art, it makes the most sense that an "artist's trademark" would be his or her own "unique style." Although answer choice F, "prime example," may appear to be correct, the author is discussing different styles of art, not just one specific piece of art. Answer choices G and H do not make sense in the context of the passage.

27. The best answer is A. As stated in the passage, the Impressionist movement was a "rebellion against Realism," which came as a result of "the rigidity and staleness some saw in the style." The rigidity and staleness are referring to the Realist style of painting, not the Impressionist style of painting. Therefore, answer choice A is correct.

28. The best answer is H. The author lists many styles that evolved from Impressionism; these details correlate best with answer choice H. Answer choice F is incorrect because Impressionism greatly expanded the range of artistic expression through the many styles of art that developed as a result of the movement. Answer choice G is incorrect because Realism was a precursor to Impressionism, not vice versa. Likewise, answer choice J is incorrect because Abstract Art and Cubism were predecessors of Impressionism, not vice versa.

29. The best answer is A. The passage states that "Postimpressionism uses form and color to reflect art in a more personal and subjective way than did its predecessor." This correlates best with answer choice A. The other answer choices are not mentioned in the passage as describing Post-Impressionism in comparison to Impressionism.

30. The best answer is J. As stated in the passage, "a rebellion against Realism arose in response to the rigidity and staleness some saw in the style." The author goes on to say that many artists began to paint in the Impressionist style because it "allowed for more creativity." These facts best support answer choice J. Although Realism could trace its roots to ancient Rome—answer choice G—this was not the reason that artists rebelled against Realism.

PASSAGE IV

31. The best answer is C. The primary focus of the passage is on gene linkage; what it is and how it works. The other answer choices are too specific and do not adequately express the overall main idea of the passage.

32. The best answer is G. The passage states that "Each hereditary unit, the *gene*, contains specific . . ." This clearly shows us that in this passage, a hereditary unit is referred to as a gene.

33. The best answer is B. The passage lists hair color, height, and susceptibility to disease all as inherited traits. Therefore, answer choice B is correct. The passage tells us that genes are found on segments of the DNA molecule, not that molecules are examples of inherited traits.

34. The best answer is J. The passage describes the fruit fly as a "*prolific* breeder, producing hundreds of offspring in a single mating." The word "prolific" means "productive and fertile," which best correlates with answer choice J. This makes the most sense in the context of the passage; if fruit flies produce so many offspring in a single mating they are clearly highly productive breeders. The other answer choices do not fit the context of the sentence as well.

35. The best answer is C. According to the first paragraph, asexual reproduction involves a single parent.

36. The best answer is F. The last paragraph of the passage indicates that current research is exploring the fact that "many factors affect the transmission of certain traits from parents to offspring." The passage goes on to state the location of genes is "but one of a multitude of determinants involved in whether or not a characteristic will be inherited."

These facts suggest that research is ongoing, and that there are still many questions regarding the transmission of genes from one generation to the next. This best correlates with answer choice F.

37. **The best answer is B.** After stating the Morgan chose to experiment on *Drosophila melanogaster*, the passage cites a reason for this choice: "The fruit fly is a prolific breeder, producing hundreds of offspring in a single mating." This best corresponds with answer choice B. Although answer choice A may appear to be correct because fruit flies have easily distinguishable chromosomes, fruit flies have only four pairs of chromosomes, not "many."

38. **The best answer is G.** As stated in paragraph 4, Morgan bred "female flies that appeared normal, but carried the mutant genes." This tells us that the female flies held the mutant genes, yet did not appear to be mutated. This statement is best supported by answer choice G. The other answer choices are not supported by the context of the passage.

39. **The best answer is D.** The last paragraph discusses the "many factors" that affect the transmission of traits, as well as the "multitude of determinants" that determine whether a characteristic will be inherited. These statements indicate that there are many components that affect genetic transmission, which is most consistent with answer choice D. The other answer choices are not supported by the information found in the last paragraph.

40. **The best answer is J.** To answer this question, a parallel must be drawn between the logic used in the passage and the question at hand. As written in the passage "the genes for body color and wing size are transmitted together from parents to offspring because they are located on the same chromosome and must be somehow linked." Therefore, if the genes for blue eyes and brown hair are found on the same chromosome, these two traits would be inherited together. Because the passage also states that combinations from both parents were present, it can be concluded that a certain number of offspring would inherit both blue eyes and brown hair, but a certain number of offspring would receive the eye and hair color of the other parent. This best correlates with answer choice J.

Science Reasoning Test Explanations

PASSAGE I

1. **The best answer is D.** Passage 1 defines macronutrients as "those nutrients required in the greatest quantity." This definition implies that plants need macronutrients more than any other element to grow and survive.

2. **The best answer is J.** Botanist 2 says, "Organic material can cause a temporary depletion of nitrogen in the soil," which suggests that when organic fertilizers are used over time, the nitrogen in the soil will be depleted. This information best supports answer choice J, because nitrogen is mentioned as a necessary nutrient.

3. **The best answer is B.** Botanist 2 states, "characteristics of organic fertilizer require application well in advance of need to ensure that the materials have broken down and can be used by the plant." The botanist is implying that the materials in organic fertilizer require a long time to be broken down and supply nutrients to the plants. This suggests the correct answer is answer choice B.

4. **The best answer is H.** The introductory material states, "Two botanists discuss whether inorganic or organic fertilizers are most optimal for plant growth"; this best supports answer choice H. The other answer choices are not supported by the passage.

5. **The best answer is D.** The best way to answer this question is by the process of elimination. Botanist 1 says, "organic fertilizer provides a naturally slow release of nutrients as the organic material breaks down in the soil, reducing the likelihood of over-fertilization." Answer choice A is true, and, therefore, incorrect. This botanist also says, "Organic fertilizers also improve soil structure in the long term," and this proves answer choice B to be incorrect. The statement "Commercial inorganic fertilizers, on the other hand, are often applied to heavily, damaging the roots of the plants," suggests that, although inorganic fertilizers damage the roots of plants, organic fertilizers do not. Answer choice C can be eliminated. Botanist 1 does not directly discuss the amount of organic fertilizer to use, so answer choice D is best.

6. **The best answer is G.** Both botanists are discussing the use of organic and inorganic fertilizers as a means to "enrich the soil and make more of these essential nutrients available." This implies that plants require essential nutrients for optimal growth, some of which can be supplied by fertilizer. Answer choice F is incorrect because plants require nutrients other than nitrogen to survive. Answer choice H is not supported by information in the passage. Answer choice J is incorrect because water is not provided by fertilization.

7. **The best answer is D.** Both botanists discuss the fact that organic fertilizers, because of their break down process, have a slow release of nutrients. This implies that inorganic fertilizers can supply nutrients to the plants at a faster rate. In addition, Botanist 2 states "Inorganic fertilizers . . . offer immediate availability of nutrients . . .," which best supports answer choice D.

PASSAGE II

8. **The best answer is H.** Table 3 shows that compared to flowers pollinated by hand, flowers pollinated by bees had a greater number of flowers reproducing after 2 years (15). The other answer choices are not supported by the data.

9. **The best answer is A.** According to Study 2 , "Two pollen containers were placed in each site: one containing 50 mg Species A pollen and one containing 50 mg Species B pollen." This means that the amount of pollen placed at each site remained consistent in the study. Answer choices B and C are not supported by the data. Answer choice D can be eliminated because this information was not discussed in the study.

10. **The best answer is F.** The information needed to answer this question is located in Table 1. The best way to answer the question is to read the answer choices and then verify if they are true according to the tables. According to Table 1, the percentage of stamen number covered with pollen is 27% for both flower species A and B, which best supports answer choice F.

11. **The best answer is B.** According to Study 2, flower Species A and B were absent at site 3. This implies that site 3 was used to determine the pollen preference when both Species A and B flowers were missing, answer choice B.

12. **The best answer is H.** According to Study 2, "The containers were left in place for 36 hours and the amount of pollen that was taken from the containers was recorded." This does not take into consideration other insects that could have removed pollen from the containers. This information best supports answer choice H. While it is true that some species of flowers were not at both sites, this was a necessary variable in the experiment, and is not considered a weakness in the experimental design.

13. **The best answer is C.** Table 2 shows the results of Study 2. By reading the table, you can see that the amount of pollen removed from the pollen dishes was greater for the flower species that was absent at each site. This implies that an important factor affecting the flower preference of bees is which flower species is available in an area. Answer choices A and D can be eliminated because the information was not discussed in Study 2.

PASSAGE III

14. **The best answer is G.** According to Figure 1, for Mammal 2, as the atmospheric pressure decreases (going to the right along the *x*-axis), the respiratory rate increases, which best supports answer choice G.

15. **The best answer is C.** Mammals 2 and 3 had the same respiratory rate when the two lines representing each mammal crossed on Figure 1. This overlap occurred when the atmospheric pressure was at approximately 0.80 atm, answer choice C.

16. **The best answer is G.** The note in Passage III says, "Larger animals typically have slower respiratory rates." Since larger animals need more oxygen, the note implies that larger animals take in more oxygen with each breath compared to smaller animals. This would allow them to take a fewer number of breaths per minute, but still get the oxygen they need to survive. In Figure 1, Mammal 4's respiratory rate is consistently lower than Mammal 2's respiratory rate. The only way Mammal 4 could take in significantly more oxygen per minute than Mammal 2 and still have a lower respiratory rate is if Mammal 4 was significantly larger than Mammal 2.

17. **The best answer is D.** If a higher respiratory rate causes mammals to have a higher metabolic rate, mammals that had a higher respiratory rate at a pressure of 1.0 atm and 0.80 atm would have a higher metabolic rate at these two pressures. According to Figure 1, Mammal 1 has a higher respiratory rate at 1.0 atm than at 0.80 atm, so eliminate answer choices B and C. Because Mammal 4's respiratory rate is slightly higher at 1.0 atm, answer choice D must be correct.

18. **The best answer is J.** Mammal 4 is the only mammal, according to Figure 1, that has a low respiratory rate at higher atmospheric pressures. According to the passage, a high respiratory rate (rapid breathing) can be a sign of distress in some animals. This suggests that Mammal 4 is more comfortable (has a low respiratory rate) at higher altitudes, answer choice J.

PASSAGE IV

19. **The best answer is B.** The question states, "the earth's atmosphere will become less effective at shielding the surface from radiation of higher frequencies." If the atmosphere is less effective at shielding higher frequency radiation, surfaces on earth will be exposed to more radiation at higher frequencies. This will cause surfaces to emit more electrons, thus, making the photoelectric effect on metals more evident, which best supports answer choice B.

20. **The best answer is H.** If the rate of the photoelectric effect is directly proportional to the surface area of the metal exposed, by exposing more metal (using a larger sheet of metal) the photoelectric effect will increase. An increase in the photoelectric effect means the charge on the sheet will increase, answer choice H. Answer choice F can be eliminated because the frequency of radiation is irrelevant to this question. The remaining answer choices are not supported by the data.

21. **The best answer is D.** Answer choice D is correct because it tests different metals and their sizes with different frequencies of radiation many times. The other answer choices only account for one variable, and, therefore, are not thorough enough. Answer choice A can be eliminated because experiments should always be repeated more than once to obtain accurate results.

22. **The best answer is J.** If a scientist wanted to measure the effect of the atmosphere on the photoelectric effect, he or she would have to test the photoelectric effect in more than one atmosphere. The only answer choice that tests the photoelectric effect in more than one atmosphere—the earth's surface and outer space—is answer choice J. Answer choice H can be eliminated because the test does not rely on the atmosphere for results.

23. **The best answer is C.** Passage IV states: "No electrons are emitted for radiation with energy frequencies below that of the threshold, as the electrons are unable to gain sufficient energy to overcome attractive forces within the metal." Therefore, the scientist must have assumed that only photons with high enough frequency will emit electrons.

24. The best answer is G. According to the results of Experiments 1 and 2, as the frequency of radiation increased, so did the electron emission (which required a high photon energy). This supports the definition of photons as finite packets of energy at various levels because higher frequency radiation caused the emission of electrons.

PASSAGE V

25. The best answer is A. Each level in the figure represents a generation of rabbits. Level 1 is the first mating, or first generation. Level 2—the second generation—represents the offspring of level 1. Level 3 represents the offspring of level 2.

26. The best answer is H. Passage V implies that each offspring will receive one allele from each parent and, "Dominant alleles are expressed whenever present but recessive alleles are expressed only when the dominant allele is absent." Out of the two parents in level 1, one parent is homozygous for at least one trait, meaning the offspring will only show dominant traits. This information best supports answer choice H.

27. The best answer is B. Figure 1 shows the different possible outcomes for the alleles of the offspring of the level 2 rabbits. The offspring will have white hair if they receive the recessive allele, *b*, from both parents. Out of the possible outcomes (16), this occurs 4 times. The correct answer is 25%, answer choice B.

28. The best answer is J. By looking at Figure 1, you can determine the ratio for the traits of the offspring of the level 2 parents. Out of the 16 possibilities, short black hair occurs 9 times, long black hair occurs 3 times, short white hair occurs 3 times, and long white hair occurs 1 time. This suggests the ratio would be 9:3:3:1, answer choice J.

29. The best answer is C. If heterozygous rabbits are bred (level 2), it is possible for the recessive traits to be visible in the immediate generation, because it is possible for an offspring to receive two recessive alleles. This is also true for future generations, making answer choice C the best answer.

PASSAGE VI

30. The best answer is G. According to Figure 1, the only pair of elements in the answer choices with a bond energy greater than 420 kJ/mol is H and O (460 kJ/mol), answer choice G.

31. The best answer is C. To answer this question, you must look at Figure 1 to determine which 3 pairs of elements have the highest bond strength. Element pairs H—F (568 kJ/mol), H—O (460 kJ/mol), and H—H (436 kJ/mol) have the highest bond strengths in the table, and are therefore, the most stable bonds, which best supports answer choice C.

32. The best answer is J. To answer this question you must determine the sum of the bond energies for each answer choice, as shown below:

F. H_2O has two H–O bonds: $(460 \times 2) = 920$ kJ/mol
G. H_2S has two H–S bonds: $(366 \times 2) = 732$ kJ/mol
H. NH_3 has three H–N bonds: $(393 \times 3) = 1{,}179$ kJ/mol
J. H_3Cl has three H–Cl bonds: $(432 \times 3) = 1{,}296$ kJ/mol

The substance with the highest sum of bond energies is H_3Cl, answer choice J.

33. The best answer is A. To answer this question correctly, you must pay attention to the headings in Figure 1. According to Group 17 in Figure 1, the group bond energy decreases as bond length increase. This information best supports answer choice A.

34. The best answer is J. To answer this question, you must look at the bond lengths between the given pairs of elements in Figure 1. The bond lengths for H–O, H–S and H–Se are 96 pm, 134 pm, and 146 pm, respectively. Therefore, the order of their bond lengths in increasing order is H–O < H–S < H–Se, or answer choice J.

PASSAGE VII

35. The best answer is C. Passage VII says, "During years with less rain, fewer bands will be formed, and the bands will be more narrow than the bands formed during years with heavier rainfall." This suggests that trees in areas with more rainfall will have more bands per year and a larger size of growth bands than trees in areas with less rainfall. Site 3's trees had an average of 20 growth bands per year that were 12 mm, which is more than Site 2's trees (15 bands per year at 4 mm).

36. The best answer is G. According to the results of the experiment, trees at Site 1 had smaller and fewer growth bands than trees at Site 2. This implies that the Site 1 trees did not grow as fast as Site 2 trees.

37. The best answer is B. According to the results of the experiment, as the average size of the growth bands increases (going down the table), the average

number of growth bands per year also increases. This causes the slope of a number versus size graph to be positive, which is indicated in answer choice B. Answer choice C can be eliminated because this graph shows that all sites had the same size of growth bands, which is not supported by the data. Likewise, the other answer choices are incorrect.

38. The best answer is H. The passage implies that trees that receive a heavier rainfall will form larger and more bands. According to the results of the experiment, Site 3 had the largest average number of growth bands per year (20) and the largest average size of growth bands (12 mm). This information supports answer choice H.

39. The best answer is A. Passage VII says that *dendrochronology* helps to determine the growth rate of trees but, "Because *dendrochronology* is not completely accurate on its own, it is often combine with a process called cross dating." This implies that cross dating is applied in order to improve the accuracy of determining the growth rate of trees.

40. The best answer is G. The question states that trees from Site 4 were found to have an average of 13 growth bands per year, in between Site 1's trees (11 bands) and Site 2's trees (15 bands). This suggests that the trees from Site 4 will have an average size of growth bands that falls between the trees of Site 1 and 2. The average size of growth bands for trees from Site 1 is 2 mm and the average size of growth bands for trees from Site 2 is 4 mm. Therefore, the trees from Site 4 can be estimated to have an average size of growth bands between 2 and 4 mm.

Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.

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